

Artifact: μ Touch: Enabling Accurate, Lightweight Self-Touch Sensing with Passive Magnets

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QUICK LINKS

Code repo	github.com/Wangmerlyn/muTouch
MagX base	github.com/dychen24/magx
Models	Release tag backup/3_dim-models-20260121 on GitHub Releases (Assets)
PCB sources	pcb/ (muTouch Altium project; legacy Magway.* filenames)

I. SCOPE

This guide describes the artifact supporting μ Touch: Enabling Accurate, Lightweight Self-Touch Sensing with Passive Magnets [1]: hardware (muTouch PCB + magnets) and software (BLE data collection, semi-supervised classifier). It targets reviewers who want to install, run, and validate the pipeline. Component specs and how they are verified: hardware specs and outputs in `pcb/README.md` (used in Section V); firmware/data-collection specs and expected signals in `Codes/read_raw_ble/README.md` (used in Sections V–VI); project overview and folder purposes (including artifact/) in `README.md`.

II. NOTATION

Paths are relative to the repo root unless stated; runtime scripts assume `Codes/` as the working directory.

III. BILL OF MATERIALS & REQUIREMENTS

A. Hardware (minimal)

- muTouch PCB (Altium project in `pcb/`; assembled board; filenames use legacy `Magway.*`). PCB design by Xiaomeng Chen.
- 1–2 passive N52 grade magnets (6–8 mm recommended).
- Host laptop: Ubuntu 20.04+ or macOS 12+, 4-core CPU, ≥ 8 GB RAM, BLE 4.0+ adapter.
- Optional: BLE USB dongle (if desktop lacks BLE).

B. Software

- Python 3.10; Conda virtual env named `muTouch` recommended.
- Git with submodules; CMake/Make only if you rebuild the C++ solver.

- Dependencies installed via `pip install -r Codes/requirements.txt`.
- LaTeX/PDF tools only needed to rebuild this document.

IV. OBTAINING THE ARTIFACT

1) Clone the repository (now public):

```
git clone --recurse-submodules \
  git@github.com:Wangmerlyn/\
  muTouch.git
# HTTPS fallback:
# git clone --recurse-submodules \
#   https://github.com/Wangmerlyn/\
#   muTouch.git
```

If you hit SSH “permission denied”, use the HTTPS command above or ensure your SSH key is added to GitHub.

2) Activate env:

```
conda create -n muTouch python=3.10
conda activate muTouch
```

You should see the shell prompt prefixed with (`muTouch`) after activation; this isolates dependencies for repeatability.

3) Install deps:

```
pip install -r Codes/requirements.txt
```

Expect a clean install without errors; rerun from the repo root if paths are not found.

- ### 4) Models:
- | | |
|-------------------------------|-----|
| snapshot | tag |
| backup/3_dim-models-20260121. | |
- Download binaries from GitHub Releases (Assets).
- ### 5) Set working directory for runtime scripts to `Codes/`:

```
cd Codes
```

V. SETUP & CONFIGURATION

1) Flash firmware:

Codes/Arduino/bleReadMultiple/bleReadMultiple.ino in
Arduino IDE;
select Feather nRF52; upload.

2) Find BLE address: python read_raw_ble/find_device.py (copy device MAC/UUID).

3) Hardcode BLE address: edit the address = ''...' line near the bottom of:

- read_raw_ble/read_sensor.py
- read_raw_ble/read_sensor_real.py
- read_raw_ble/read_sensor_real_classifier.py (if used)

4) Calibration capture (run inside Codes/):

```
python read_raw_ble/read_sensor.py
```

Do a brief figure-8 motion away from metal surfaces;
CSVs are saved under datasets/.

5) Offsets/scales: place the latest files named offset-*.npz and scale-*.npz in calibration_files/. The scripts automatically load the newest files with those prefixes.

6) Models: ensure read_raw_ble/models/ holds the downloaded checkpoint set if you need pretrained clas- sifiers.

VI. RUNNING THE ARTIFACT

A. Data capture

Captures raw magnetometer streams over BLE with the latest
calibration, writing timestamped CSVs under datasets/.

```
python read_raw_ble/read_sensor_real.py
```

B. Real-time classification

Runs the live classifier; expect gesture labels printed to console
and logged in datasets/.

```
python read_raw_ble/read_sensor_real_\  
classifier.py
```

Ensure the script uses the latest offset-*,
scale-*, and model files (picked automatically
if they sit in calibration_files/ and
read_raw_ble/models/).

C. Expected outcomes

- Face-touching: $\approx 93\%$ accuracy (8 gestures) with 3 s fine-tuning/user.
- Scratch detection: $\approx 95\%$ accuracy (9 gestures) with 3 s fine-tuning/user.
- Real-time loop maintains $>30\text{Hz}$ inference on a laptop CPU.

VII. REPRODUCIBILITY CHECKLIST

- **Hardware reproducible:** PCB sources + BOM (muTouch; files named Magway.* for compatibility) included.
- **Software reproducible:** All scripts + TS2Vec submodule; pinned deps in Codes/requirements.txt.
- **Data:** Calibration and small demo runs can be generated locally; full datasets are participant-specific and not included.
- **Pretrained models:** Provided via GitHub tag backup/3_dim-models-20260121.

VIII. TROUBLESHOOTING

- BLE not found: retry find_device.py; check power and pairing blocks; use BLE dongle.
- Drifting predictions: recalibrate sensors; ensure distance from large metal; re-run offset/scale.
- Import errors: confirm submodule init (git submodule update --init --recursive) and Python path from repo root.

IX. FOR COMPLETE INSTRUCTIONS

Due to page limits, this PDF is a concise checklist. For full, continuously updated steps and tips, please refer to the repository README.md and the Markdown guide (best viewed via GitHub preview): artifact/README.md. Use those as the authoritative references for reproduction and further code tweaks.

X. NOTES ON PRIOR WORK

The project builds on MagX (MobiCom'21) codebase for magnetic sensing [2]. This artifact extends it to self-touch sensing and includes updated PCB by Xiaomeng Chen.

REFERENCES

- [1] S. Wang, K. Li, J. Huang, J. Wang, C. Zhang, A. Sample, and D. Chen, " μ touch: Enabling accurate, lightweight self-touch sensing with passive magnets," in *Proceedings of the IEEE International Conference on Pervasive Computing and Communications (PerCom)*, Pisa, Italy, Mar. 2026, to appear.
- [2] D. Chen, M. Wang, C. He, Q. Luo, Y. Iravantchi, A. Sample, K. G. Shin, and X. Wang, "Magx: Wearable, untethered hands tracking with passive magnets," in *Proceedings of the 27th Annual International Conference on Mobile Computing and Networking (MobiCom '21)*, 2021, pp. 269–282.